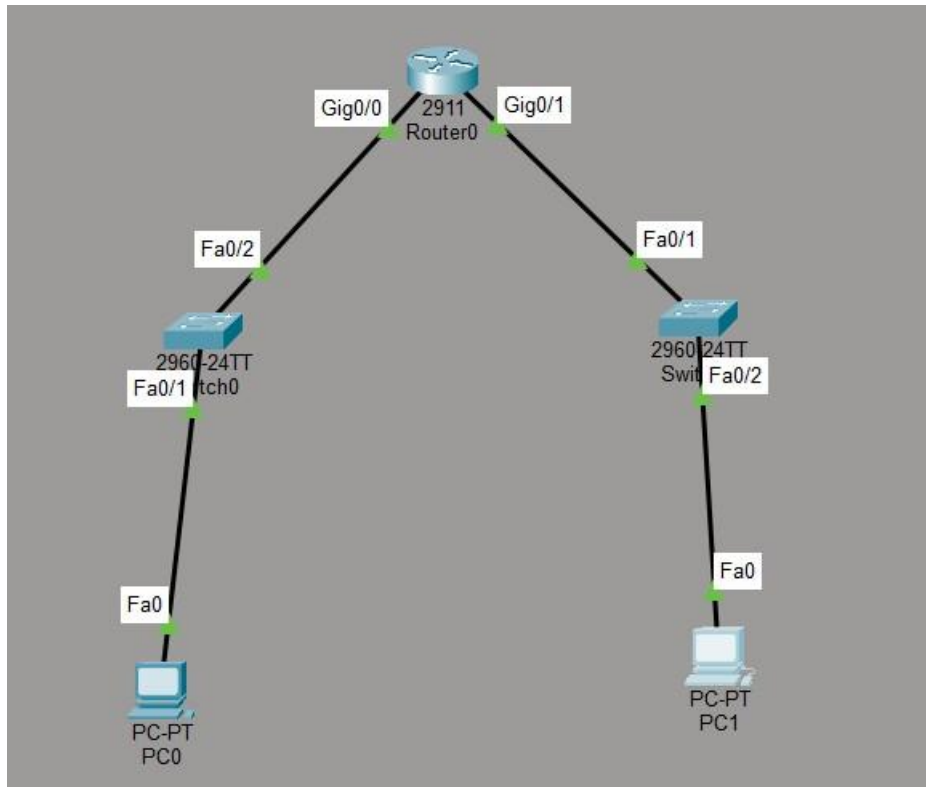


Tasks'

1. In Cisco Packet Tracer, set up a simple network with a router, one internal PC, and one external PC. Assign private IP addresses to the internal PC and public IP addresses to the external PC, then configure static NAT on the router to allow the internal PC to communicate with the external PC.



```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g 0/1
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip nat inside source static 192.168.1.10 203.0.113.2
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
---  ---                ---              ---              ---
---  203.0.113.2         192.168.1.10     ---              ---
```

```

C:\>ping 203.0.113.10

Pinging 203.0.113.10 with 32 bytes of data:

Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time=1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127

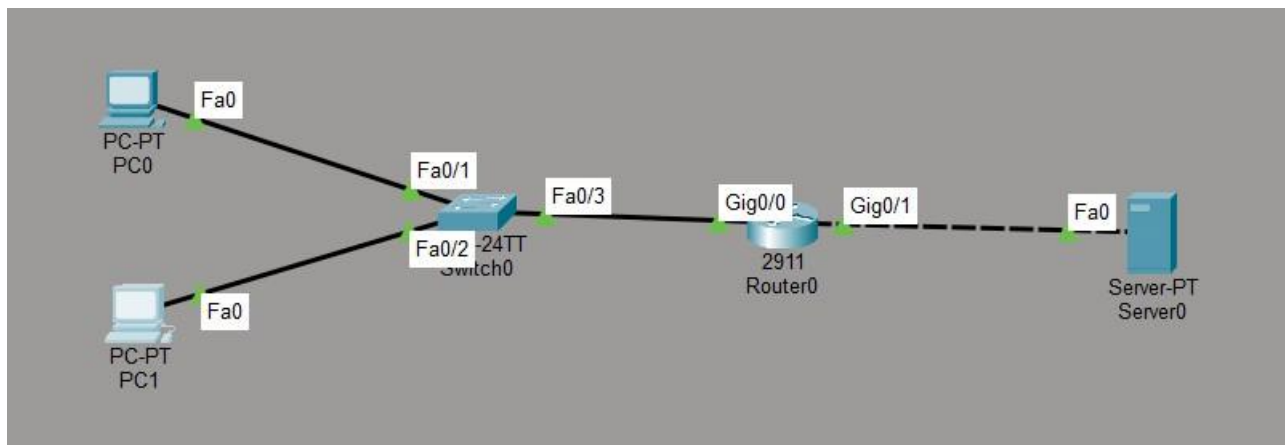
Ping statistics for 203.0.113.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

2. Configure Port Address Translation (PAT) on a router in Packet Tracer so that two internal PCs with private IPs can access a simulated public web server using a single public IP address.

Hint: Use the 'ip nat inside source list' command with 'overload' to enable PAT.



```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g 0/1
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

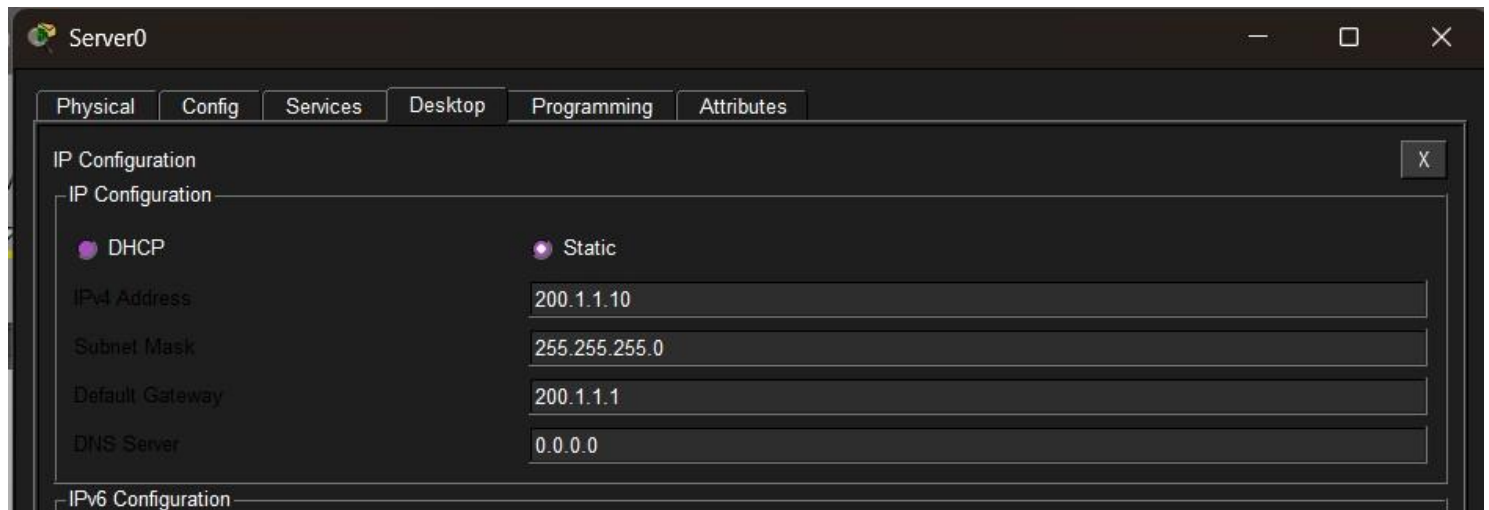
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface gigabitEthernet 0/1 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
tcp 200.1.1.1:1025        192.168.1.10:1025 200.1.1.10:80      200.1.1.10:80

Router#

```



PC 0:

```

C:\>ping 200.1.1.10

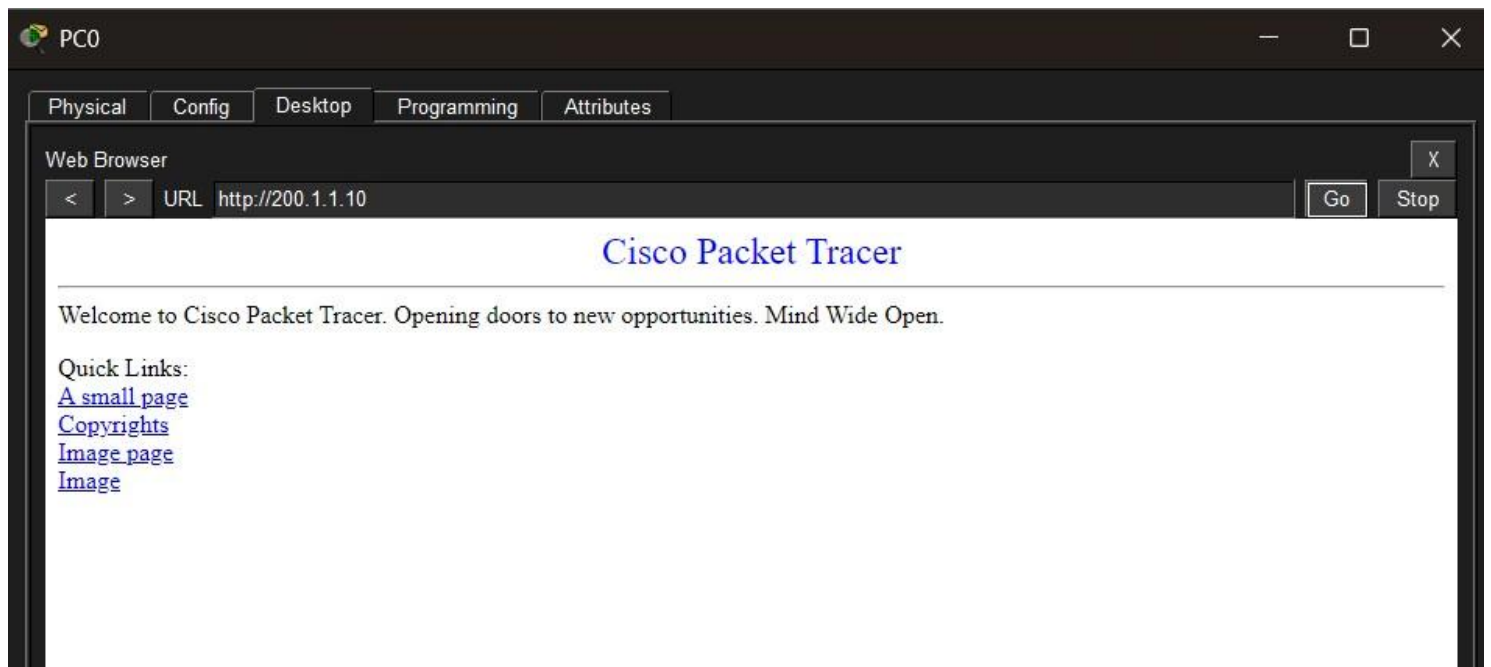
Pinging 200.1.1.10 with 32 bytes of data:

Reply from 200.1.1.10: bytes=32 time<1ms TTL=127
Reply from 200.1.1.10: bytes=32 time=1ms TTL=127
Reply from 200.1.1.10: bytes=32 time<1ms TTL=127
Reply from 200.1.1.10: bytes=32 time<1ms TTL=127

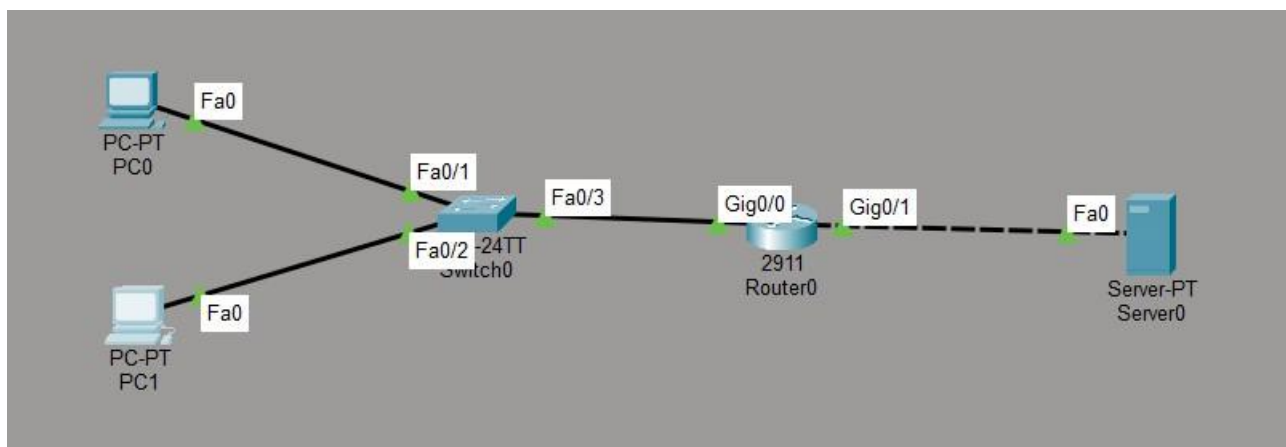
Ping statistics for 200.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```



3. Test your NAT configuration by sending a ping from the internal PC to the external PC and capture the NAT translation table on the router to verify the address mapping.



```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g 0/1
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#ip nat outside
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface gigabitEthernet 0/1 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
tcp  200.1.1.1:1025        192.168.1.10:1025 200.1.1.10:80      200.1.1.10:80

Router#show ip nat statistics
Total translations: 6 (0 static, 6 dynamic, 6 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 25 Misses: 14
Expired translations: 8
Dynamic mappings:
Router#

```

PC 0 Ping

```

C:\>ping 200.1.1.10

Pinging 200.1.1.10 with 32 bytes of data:

Reply from 200.1.1.10: bytes=32 time<1ms TTL=127
Reply from 200.1.1.10: bytes=32 time=1ms TTL=127
Reply from 200.1.1.10: bytes=32 time<1ms TTL=127
Reply from 200.1.1.10: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

PC 1 Ping

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.1.1.10

Pinging 200.1.1.10 with 32 bytes of data:

Reply from 200.1.1.10: bytes=32 time<1ms TTL=127
Reply from 200.1.1.10: bytes=32 time<1ms TTL=127
Reply from 200.1.1.10: bytes=32 time<1ms TTL=127
Reply from 200.1.1.10: bytes=32 time<1ms TTL=127

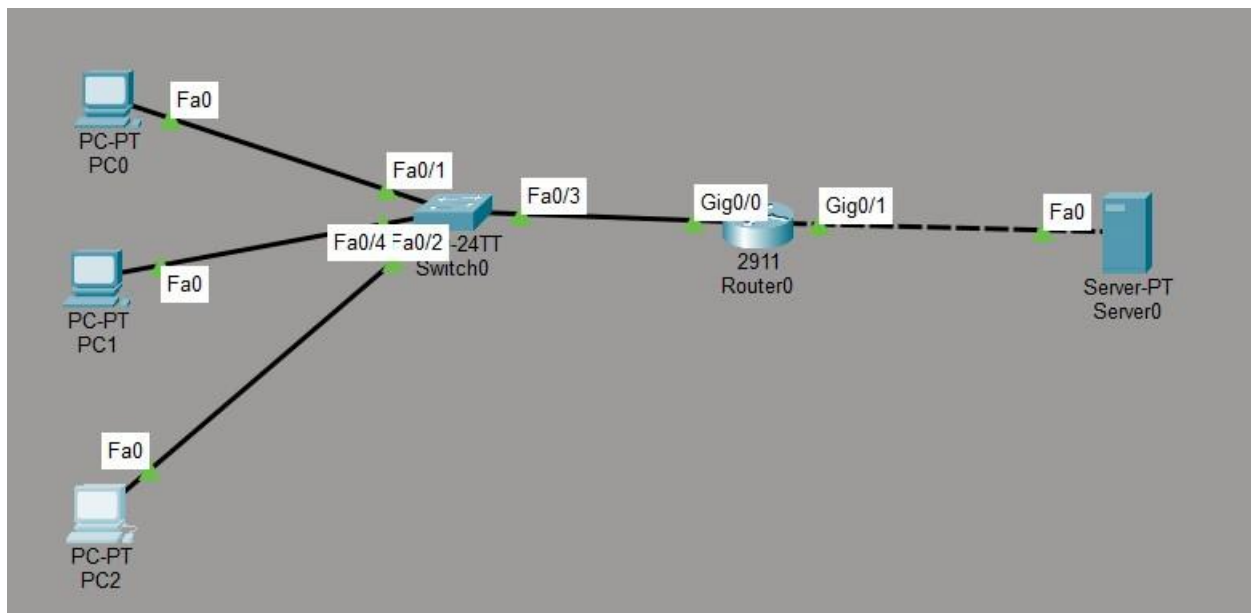
Ping statistics for 200.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

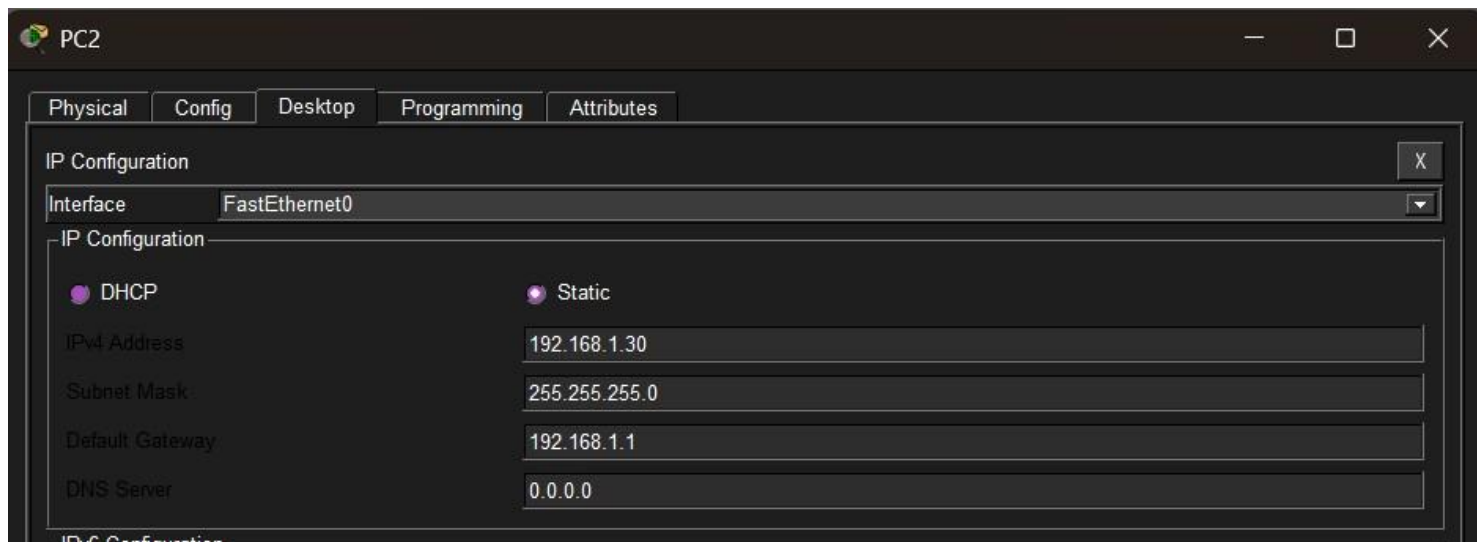
```

The NAT translation table confirms that the private IP 192.168.1.10 is translated to the public IP 200.1.1.1 using PAT. The external web server 200.1.1.10 is accessed through HTTP port 80. The NAT statistics also show successful translated traffic, confirming that PAT is functioning correctly

4. **Modify your Packet Tracer topology to add a third internal PC. Update your NAT or PAT configuration so all three internal PCs can access the external network, and verify that each translation appears in the router's NAT table.**



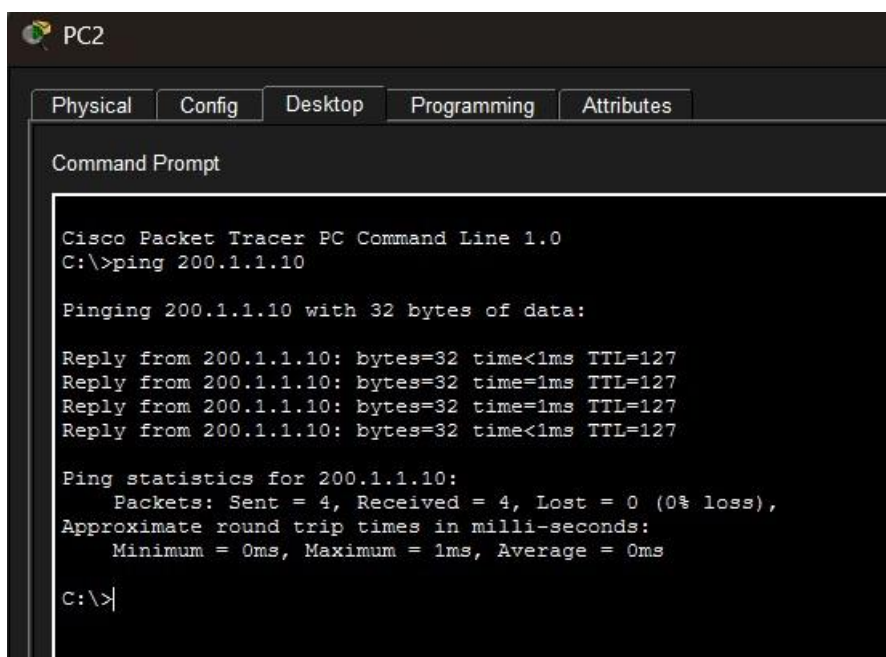
PC 2 :



Router

```
Router#
Router#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 200.1.1.1:1        192.168.1.30:1    200.1.1.10:1       200.1.1.10:1
icmp 200.1.1.1:2        192.168.1.30:2    200.1.1.10:2       200.1.1.10:2
icmp 200.1.1.1:3        192.168.1.30:3    200.1.1.10:3       200.1.1.10:3
icmp 200.1.1.1:4        192.168.1.30:4    200.1.1.10:4       200.1.1.10:4
tcp  200.1.1.1:1025     192.168.1.10:1025 200.1.1.10:80      200.1.1.10:80
tcp  200.1.1.1:1026     192.168.1.10:1026 200.1.1.10:80      200.1.1.10:80
Router#
```

PC 2 Ping



5. Explain in your own words how NAT and PAT help apps like Zomato or Flipkart serve millions of users with limited public IP addresses. Give two real-world scenarios where incorrect NAT/PAT setup could break connectivity for users.



6. NAT and PAT in real-world apps

Apps like Zomato and Flipkart have millions of users, but they cannot give every device its own public IPv4 address because public IP addresses are limited.

NAT (Network Address Translation) solves this by converting private IP addresses into public IP addresses when devices communicate with the internet.

PAT (Port Address Translation) goes one step further. It allows many devices to share the same public IP address by using different port numbers to keep their connections separate.

For example:

User 1: 192.168.1.10:5001 —┐

User 2: 192.168.1.11:5002 —┴→ Public IP: 49.x.x.x → Internet User 3: 192.168.1.12:5003 —┐

So, thousands of users can access websites and apps while using a much smaller number of public IP addresses. This helps save IPv4 addresses and makes network management easier.

Two real-world problems caused by incorrect NAT/PAT

Online payment or login failure

Suppose a user is using Zomato and the payment service requires a connection back to the user's network. If NAT is incorrectly configured or required ports are not translated properly, the connection can fail. The user might see a payment timeout or failed transaction.

Apps or services become unreachable

Suppose a company configures PAT incorrectly on its router. If the wrong interface is marked as ip nat inside or ip nat outside, users may have internet connectivity problems. For example, Flipkart pages might not load, APIs could time out, or users could be unable to log in.